

The comet assay with 8 mouse organs: results with 39 currently used food additives.

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We determined the genotoxicity of 39 chemicals currently in use as food additives. They fell into six categories-dyes, color fixatives and preservatives, preservatives, antioxidants, fungicides, and sweeteners. We tested groups of four male ddY mice once orally with each additive at up to 0.5xLD(50) or the limit dose (2000mg/kg) and performed the comet assay on the glandular stomach, colon, liver, kidney, urinary bladder, lung, brain, and bone marrow 3 and 24h after treatment. Of all the additives, dyes were the most genotoxic. Amaranth, Allura Red, New Coccine, Tartrazine, Erythrosine, Phloxine, and Rose Bengal induced dose-related DNA damage in the glandular stomach, colon, and/or urinary bladder. All seven dyes induced DNA damage in the gastrointestinal organs at a low dose (10 or 100mg/kg). Among them, Amaranth, Allura Red, New Coccine, and Tartrazine induced DNA damage in the colon at close to the acceptable daily intakes (ADIs). Two antioxidants (butylated hydroxyanisole (BHA) and butylated hydroxytoluene (BHT)), three fungicides (biphenyl, sodium o-phenylphenol, and thiabendazole), and four sweeteners (sodium cyclamate, saccharin, sodium saccharin, and sucralose) also induced DNA damage in gastrointestinal organs. Based on these results, we believe that more extensive assessment of food additives in current use is warranted.